PROJECT INITIATION REPORT (PIR)

Edinburgh Napier University
School of Engineering
Module: SOC09109 – Group Project

Project Title: PHOBOT – Android-Controlled Kobuki Mobile Robot

Team Name: Team_Phobot

Submission Date: 1st MAY 2026

Team Members:

1. Al Sifat Siddiki – Lead Developer & Integration
2. Atif Azeem – Project Manager
3. Zed Ukenye – Movement Control Specialist
4. Hafiz Abd Ur Rehman – Audio Processing & QA Lead
5. Talha Ali – Computer Vision & Object Tracking
6. Jackson Dobbie – Android Support & System Testing

Client: **Brian Davison**

Email: b.davison@napier.ac.uk

Sponser: **Petros Karadimas**

Executive Summary:

PHOBOT is an Android-controlled Kobuki mobile robot developed for the SOC09109 Group Project module. The system integrates Android-based control, movement control, and audio interaction. The project follows agile methodology with sprint-based development and regular progression meetings. The primary assessment emphasis is on functionality and code quality, supported by professional documentation.

1. Purpose and Expected Benefits

The purpose of this project is to design and implement a stable Android-controlled robotic system using the Kobuki mobile robot platform. The project enables practical experience in robotics integration, mobile application development, and agile team management.

Expected Benefits:

- Practical robotics and embedded systems experience
- Stronger agile project management skills
- Demonstration of reliable Android-to-robot communication
- Improved system integration capability

2. Expected Cost and Duration

The PHOBOT project will be delivered by the 1st of May 2026, in alignment with the SOC09109 module assessment schedule. The project timeline is primarily constrained by the academic semester duration rather than client-imposed deadlines.

Development will follow an Agile sprint structure:

- **Sprint 1 (2 weeks): Hardware setup and USB communication validation**
- **Sprint 2 (2 weeks): Core movement control implementation**
- **Sprint 3 (2 weeks): Android application integration**
- **Sprint 4 (2 weeks): Audio processing integration**
- **Sprint 5 (2 weeks): System integration and testing**
- **Final Week: System refinement and demonstration preparation**

Main Milestones

- **PIR approval**
- **Stable USB communication established**
- **Movement control functional**
- **Android integration completed**
- **Audio integration completed**
- **Full system integration achieved**
- **Final live demonstration**

Contingency Time

A one-week contingency buffer has been reserved prior to the final demonstration to address unforeseen integration issues, hardware instability, or performance optimisation needs.

Source of Constraints

The primary constraints are:

- **Module submission deadline (1st May 2026)**
- **Hardware availability (Kobuki platform)**
- **Academic workload across other modules**

The deadline is fixed due to assessment requirements and is therefore non-negotiable. However, feature scope remains flexible through MoSCoW prioritisation, ensuring that core deliverables are completed even if stretch goals are deferred.

Financial Considerations

There is no allocated financial budget for this academic project. Hardware (Kobuki robot) and development tools (Android Studio) are provided by the university. No additional licensing costs are anticipated.

3. Requirements and Quality Expectations (MoSCoW)

The following major requirements define the minimum success criteria of the project and are prioritised using the MoSCoW method.

Must Have

- **Reliable Android-to-Kobuki movement control**
- **Stable USB communication between device and robot**
- **Basic audio interaction capability**
- **Fully integrated system demonstration**

Quality Expectations for Must-Have Requirements

To be considered successful, the system must meet the following measurable criteria:

- **Stable operation for at least 3 - 5 continuous minutes during live demonstration without crash**
- **Accurate execution of movement commands (forward, backward, left, right, stop)**
- **No loss of USB communication during standard operation**

Should Have

- **Improved user interface design**
- **Error handling and user feedback mechanisms**

Could Have (If Time Allows)

- **Bluetooth connectivity**
- **Wi-Fi connectivity**
- **Basic object/person following using camera**

- **Microphone-based movement decisions**
- **Speech interaction features**

Won't Have

- **Advanced autonomous AI navigation**
- **Commercial deployment features**

4. Scope Definition In

Scope:

• Android application development (Android Studio)

• Kobuki robot movement control integration

• Audio processing integration

• Basic computer vision following behaviour (if time permits)

• Testing and final demonstration Out of Scope:

• High-precision positioning systems

• Full ROS implementation (optional inspiration only)

• Commercial production system

5. Stakeholder Map

• Brian Davison – Client (High Interest / High Influence)

• Sponsor – **Petros Karadimas**

• SOC09109 Module Leader – Academic Supervisor

• Team_Phobot – Development Team

• Peer Review – Academic Reviewer

6. Work Breakdown Structure (WBS)

1. Initiation – PIR preparation and approval

2. Planning – Requirement confirmation and sprint planning

3. Development – Android, Movement, Audio modules

4. Integration – Hardware & communication integration

5. Testing – Unit and integration testing

6. Demonstration & Submission

7. Risk Analysis

• Hardware malfunction – Mitigation: Early hardware testing

• Communication instability – Mitigation: Incremental integration

• Time constraints – Mitigation: Prioritise core features first

• Integration complexity – Mitigation: Modular architecture

8. Demonstration Plan

Primary format: Live demonstration during progression meeting. Backup:
Recorded demo in case of technical issues.

9. Early Project Strategy

During the first two weeks, the team will prioritise hardware testing, confirm stable communication, and establish a clean modular architecture to reduce integration risks later in the project.

Appendix A – Initial Follow-Up Items

The following risks have been identified during the initiation phase of the PHOBOT project. These items will be reviewed weekly during sprint meetings and updated where necessary.

ID	Risk Description	Impact	Mitigation Strategy	Owner
R1	Kobuki hardware malfunction or instability	High	Conduct early hardware testing and confirm base functionality during Sprint 1	Zed Ukenye
R2	USB communication instability between Android device and robot	High	Establish and validate stable USB communication before integrating additional features	Al Sifat Siddiki
R3	Integration complexity between movement, audio, and vision modules	High	Use modular architecture and perform incremental integration testing	Entire Team
R4	Delay due to stretch features (computer vision / object tracking)	Medium	Prioritise Must-Have features first and implement vision only after core stability	Talha Ali
R5	Android compatibility or application testing issues	Medium	Conduct early testing on available devices and perform continuous system testing	Jackson Dobbie
R6	Time constraints due to academic workload	Medium	Maintain sprint discipline and weekly progress tracking under Azeem project manager supervision	Atif
R7	Limited availability of reference material or documentation for Kobuki platform	Medium	Identify and review available technical documentation early; consult academic resources or client if clarification is required	Al Sifat Siddiki

An additional risk related to the availability of reference material was added following client feedback.

Appendix B – Peer Review

Date of Review: 13 February 2026
Reviewer: Amber Brautigam-Fox
Method: Email feedback

Amber provided positive feedback on the overall structure and clarity of the PIR. She highlighted that the project scope is clearly defined and easy to follow. She also noted that the sprint structure, defined goals, inclusion of contingency time, and early testing approach are strong aspects of the planning.
Additionally, she commented that the risk register is realistic and demonstrates an honest evaluation of potential project challenges.
She suggested that the stakeholder interests and communication plan could be expanded further and that the Executive Summary could include more measurable outcomes to strengthen clarity.
The team will review these suggestions and make appropriate updates before final submission.

Appendix C – Client Confirmation

Email Confirmation – Brian Davison

Hi Atif

The document looks good so I am happy to sign it off. I think you may need to add one or more risk items related to the availability of reference material, but apart from that it's looking good. I think the MoSCoW prioritisation is sensible - it keeps the basic scope nice and compact.

Brian